STAT 7900: Special Topic: Causal Analysis



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STAT 7900

Three Kinds of questions

Descriptive

What is happening?

Example: How does blood pressure differ between smokers and non-smokers?

Predictive

What will happen?

Example: Can age, BMI or smoking predict future blood pressure?

Causal

What would happen if we intervened?

Example: Would smoking cessation lower blood pressure?

Association is not causation

Observed association

$$E[Y \mid T = 1] - E[Y \mid T = 0]$$

$\neq$ **in general**

Causal effect

$$E[Y(1)] - E[Y(0)]$$

Why can they differ?

- Confounding: treated and untreated groups may differ before treatment.
- Selection and treatment decisions can create strong associations without causal effects.
- Simpson's paradox: aggregate and within-group comparisons can point in different directions.

Key idea: correlation can be a mixture of causal and noncausal association.

Formulating a causal question

Five ingredients of a causal question

1	2	3	4	5
Population	**Treatment**	**Comparator**	**Outcome**	**Time**
Who?	What intervention?	Compared with what?	What endpoint?	Over what horizon?

Potential Outcomes

World 1

Treatment Received

A = 1

Outcome: Y(1)

World 2

No treatment

A = 0

Outcome: Y(0)

Fundamental Problem of Causal Inference

If $T_i = 1$

Observed: $Y_i(1)$

Missing: $Y_i(0)$

FACTUAL

$\longleftrightarrow$

COUNTERFACTUAL

If $T_i = 0$

Observed: $Y_i(0)$

Missing: $Y_i(1)$

Causal inference is therefore a structured missing-data problem.

What causal effects we are trying to estimate

ATE: $E[Y(1) - Y(0)]$ **Average effect in the target population**

ATT

$E[Y(1) - Y(0) \mid T = 1]$

Effect among those treated

ATC

$E[Y(1) - Y(0) \mid T = 0]$

Effect among those untreated

CATE(x)

$E[Y(1) - Y(0) \mid X = x]$

Effect in a covariate-defined subgroup

"Does the treatment work?" is incomplete until we specify for whom and which estimand.

Causal Assumptions

Consistency

Observed Y equals the potential outcome under the treatment actually received.

Exchangeability

After conditioning on X, treatment is independent of the potential outcomes.

Positivity

Within relevant X strata, each treatment option has positive probability.

SUTVA / no interference

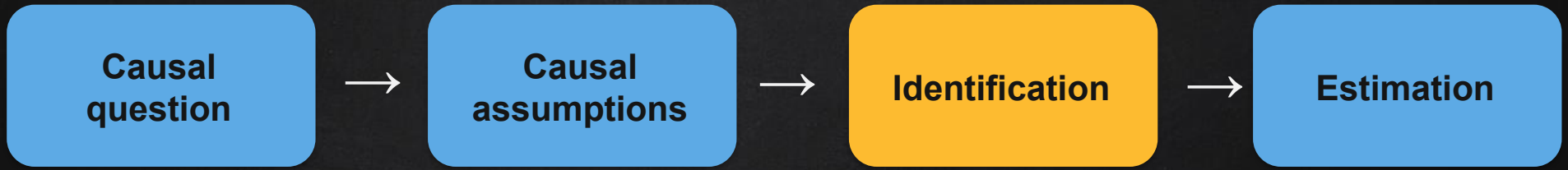
One unit's treatment does not alter another unit's outcome; treatment is well defined.

These assumptions are not fully testable from the observed data alone.

Bridge to today

A statistical model can always produce an estimate.

The causal question is whether the assumptions justify interpreting it as an effect.



Last lecture: “Assume exchangeability after conditioning on X.”

Today’s question: Which variables belong in X — and which variables should we NOT adjust for?

→ Directed Acyclic Graphs (DAGs)

STAT 7900 Quiz 1



Today: Assumptions to Adjustment

- Why causal graphs? What a DAG does, and does not, claim.
- Nodes, arrows, parents, descendants, and paths.
- The three fundamental structures: chain, fork, collider.
- d-separation: when conditioning blocks or opens a path.
- Backdoor paths and valid covariate-adjustment sets.
- Why “control for everything” can create bias.
- R coding lab: confounding, mediation, and collider bias.

Conditional Exchangeability

$$Y(t) \perp T \mid X$$

Within levels of X, treatment assignment carries no additional information about the potential outcomes.

But what belongs in X?

Dangerous Instinct: Adjust for Everything

A common regression instinct:

Put every available covariate in the model. More adjustment must be safer.

That is false for causal inference.

Some variables remove bias.
Some variables remove part of the effect you want.
Some variables CREATE bias when conditioned on.

Why use a graph

- Potential outcomes define the causal estimand precisely.
- Graphs encode assumptions about how variables generate one another.
- Those assumptions imply which associations should disappear after conditioning.
- The graph helps us decide whether a causal effect is identifiable, and what to adjust for.

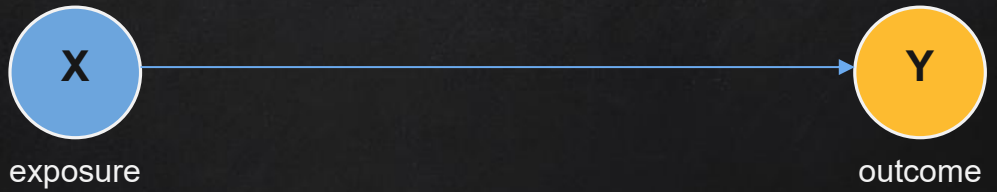
Potential outcomes tell us WHAT effect means; DAGs help organize WHY identification may be possible.

DAG

D : Directed, **A : Acyclic,** **G : Graph.**

Directed: arrows have orientation.
Acyclic: following arrows can never bring you back to the same node.
Graph: nodes connected by edges.

A causal arrow is a scientific claim.



Means: intervening on X could change Y through a direct causal mechanism represented by this edge.

DAG

No arrow is also an assumption.

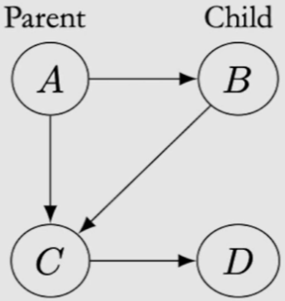


asserts that the graph does not include a direct causal effect of X on Z, given the variables represented.

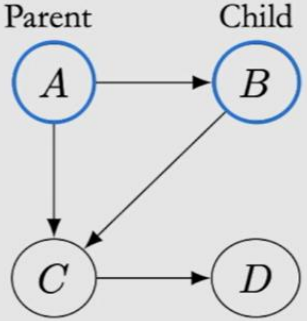
Graphical conclusions are only as credible as the causal assumptions used to draw the graph.

DAG: Directed Acyclic Graph

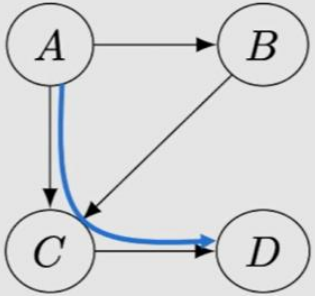
Directed Graph



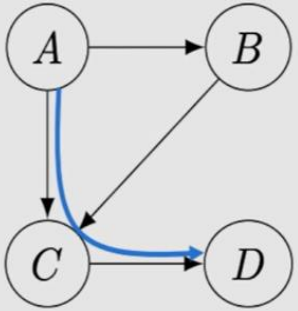
Adjacent



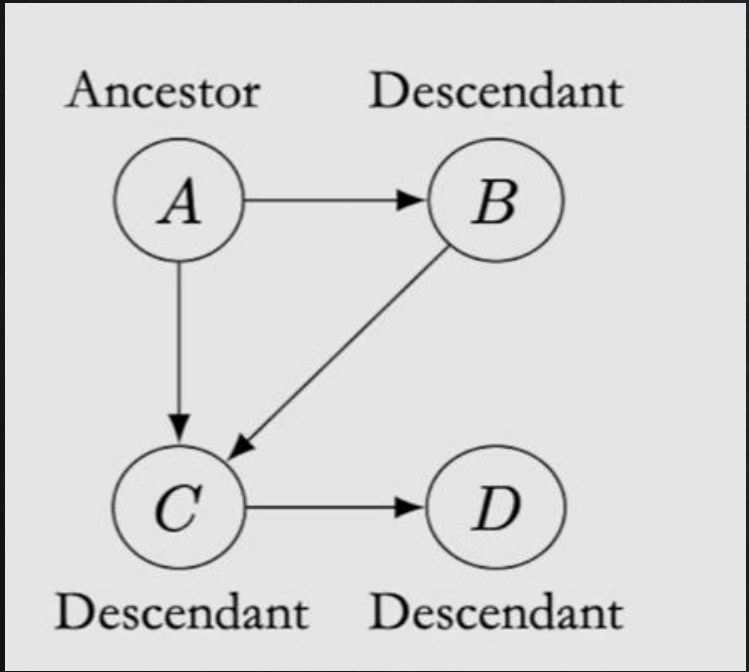
Path

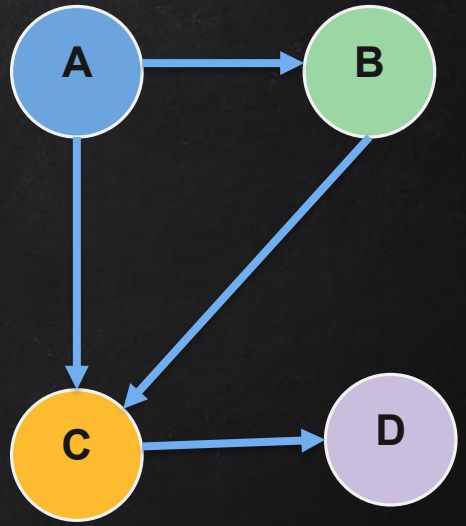


Directed Path



DAG





Ancestors of C: A,B

Descendents of A: B, C, D

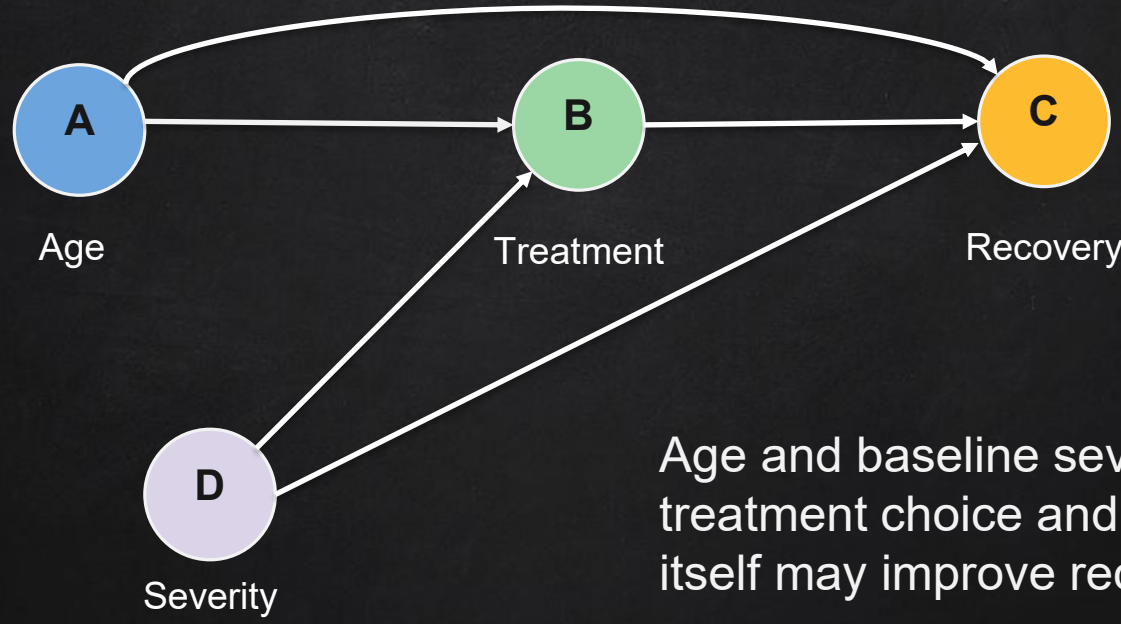
Ancestors mean there is a directed path onto the node.

DAG: Directed Acyclic Graph



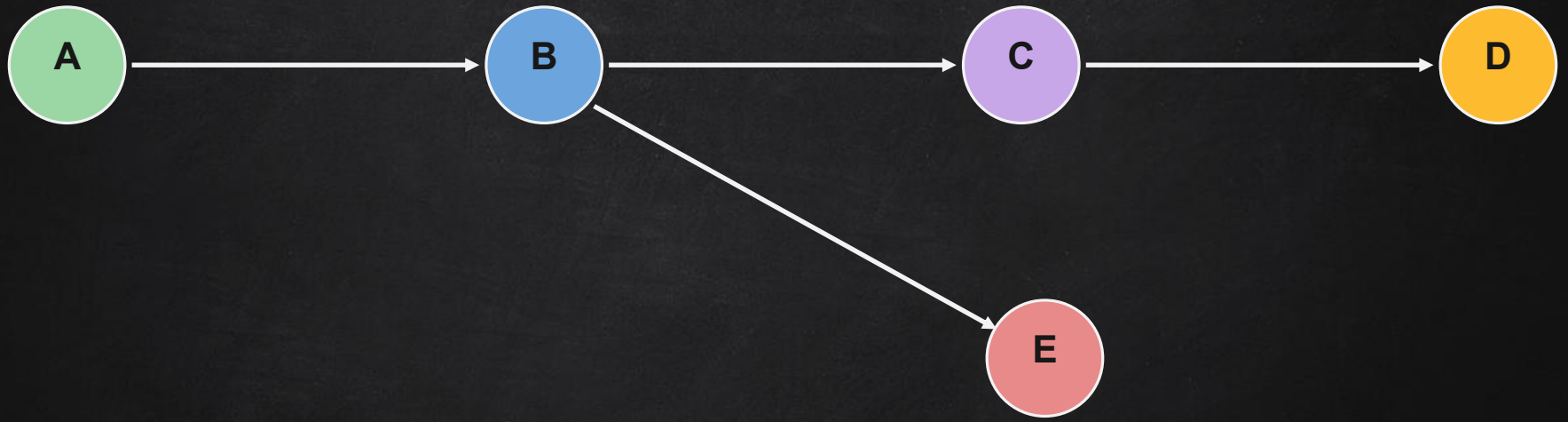
DAG: Directed Acyclic Graph

MEDICAL EXAMPLE



Age and baseline severity influence both treatment choice and recovery; treatment itself may improve recovery.

DAG: Directed Acyclic Graph



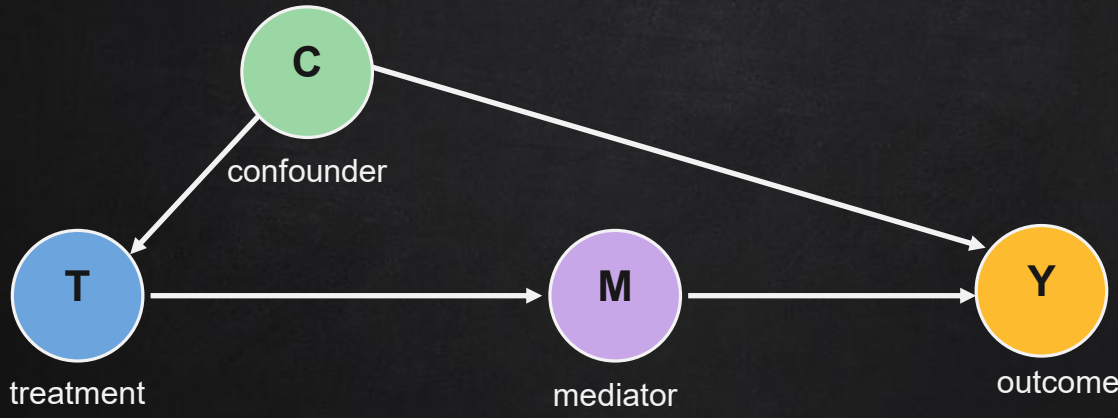
1. Parents of C?
2. 3. Ancestors of D?

2. Descendants of B?
4. Is E an ancestor of D?

Paths

A path is a sequence of adjacent edges, regardless of arrow directions.

A directed path is where every arrow must point along the path.



Causal Path:

$T \rightarrow M \rightarrow Y$

Non-causal Path:

$T \leftarrow C \rightarrow Y$

Three Building Blocks

CHAIN

$X \rightarrow Z \rightarrow Y$

Transmission of association

FORK

$X \leftarrow Z \rightarrow Y$

Common cause confounding

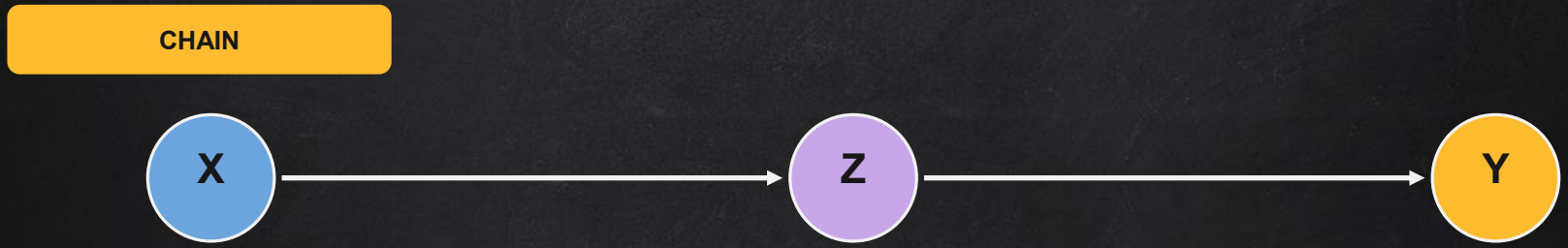
COLLIDER

$X \rightarrow Z \leftarrow Y$

Common effect selection

Conditioning behaves differently in the collider.

Structure 1: Chain



X influences Y through Z.

Without conditioning: path open.

$X \perp Y \mid Z$ **(for this path)**

X and Y can be associated.

Intuition: once Z is known, X supplies no additional information about Y through this chain.

Structure 2: Fork / Common Cause

FORK



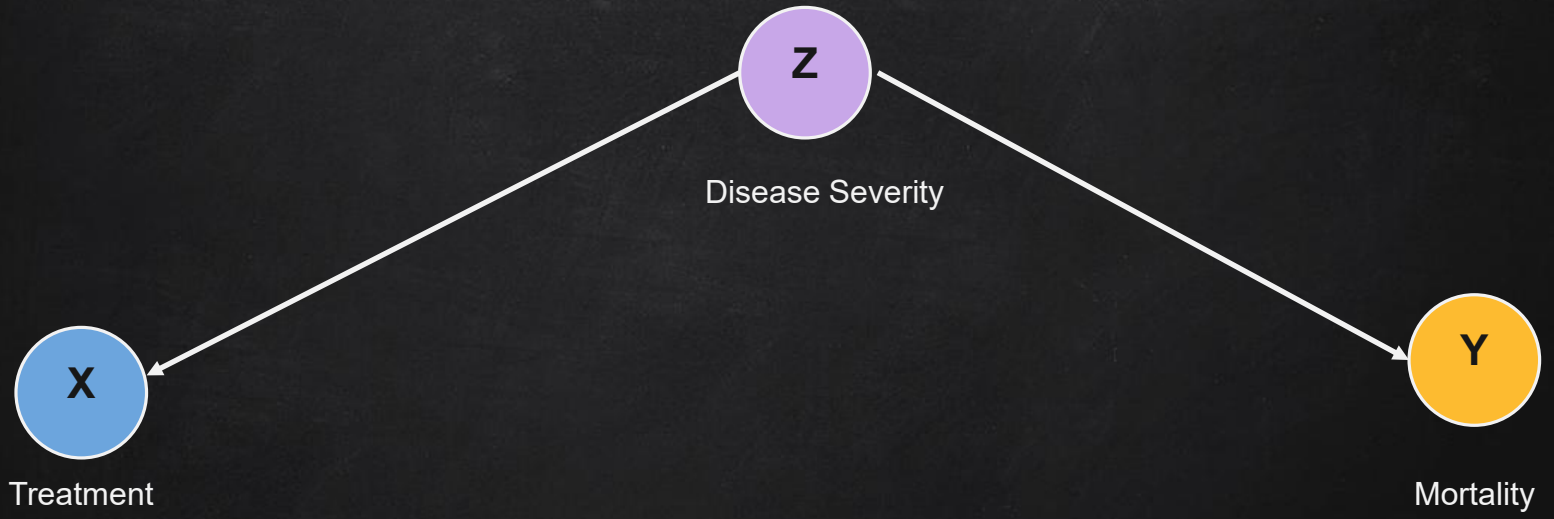
A shared cause can make X and Y associated even when X does not cause Y.

Stratifying on the common cause removes the spurious pathway through Z.

$X \perp Y \mid Z$ **(for this fork)**

Structure 2: Fork / Common Cause

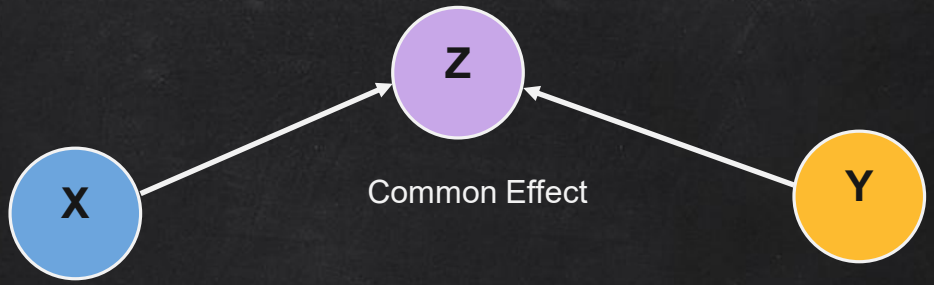
FORK



Severe patients are more likely to receive treatment AND more likely to die.

Structure 3: Collider / Common Effect

COLLIDER



Knowing X tells us nothing about Y merely because they share a consequence.

Without conditioning on Z, the path is closed.

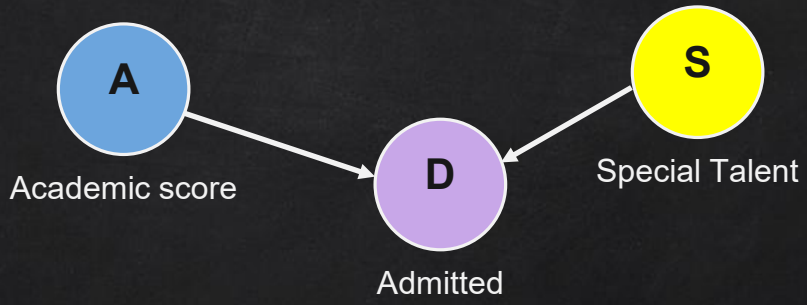
After conditioning on Z, X and Y can be associated.

This is collider bias / selection bias.

Structure 3: Collider / Common Effect

Example

Admissions to an elite program depend on BOTH academic score and special talent.



Among admitted students only, a lower academic score may imply greater special talent.

Chains, forks, colliders: One table to remember

Structure	Unconditioned	Condition on middle
$X \rightarrow Z \rightarrow Y$	OPEN	BLOCKED
$X \leftarrow Z \rightarrow Y$	OPEN	BLOCKED
$X \rightarrow Z \leftarrow Y$	BLOCKED	OPEN

Collider = the exception: conditioning OPENS instead of closes.

Open or Closed

For each path, say whether it is OPEN before conditioning and after conditioning on Z.

A. $X \rightarrow Z \rightarrow Y$

B. $X \leftarrow Z \rightarrow Y$

C. $X \rightarrow Z \leftarrow Y$

Open or Closed

For each path, say whether it is OPEN before conditioning and after conditioning on Z.

A. $X \rightarrow Z \rightarrow Y$

B. $X \leftarrow Z \rightarrow Y$

C. $X \rightarrow Z \leftarrow Y$

A. Chain
B. Fork
C. Collider

d-Separation

Two variables are d-separated by a set S when EVERY path between them is blocked after conditioning on S.

$$X \perp_D Y \mid S$$

Graphical separation → conditional independence under the causal Markov property.

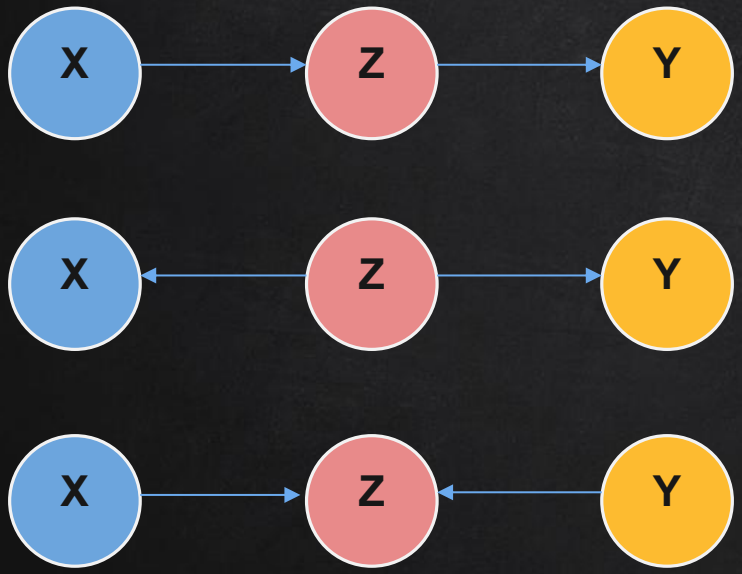
d-Separation

RULES

- A chain $X \rightarrow Z \rightarrow Y$ is blocked if we condition on Z.
- A fork $X \leftarrow Z \rightarrow Y$ is blocked if we condition on Z.
- A collider $X \rightarrow Z \leftarrow Y$ is blocked when we do NOT condition on Z or its descendants.
- A collider path becomes open if we condition on the collider, or on one of its descendants.

To d-separate X and Y, EVERY path between them must be blocked.

d-Separation



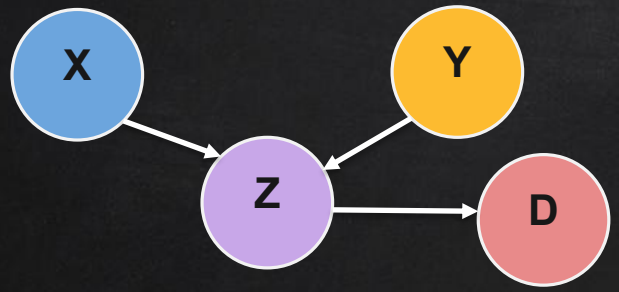
$Z \in S$, the chain is blocked by conditioning on Z.

$Z \in S$, the backdoor fork is blocked by conditioning on Z.

$Z \notin S$, the collider path is blocked automatically.

d-Separation

Conditioning on a DESCENDANT of a collider can open the path.



Conditioning on D gives information about Z, which activates the collider path.

d-Separation

Suppose hospital admission H is a collider.

Disease Severity $\rightarrow H \leftarrow$ Insurance Status.

Now suppose “received ICU bed” is caused by H.

$H \rightarrow$ ICU Bed.

Selecting on ICU-bed recipients can indirectly select on H and induce collider bias.

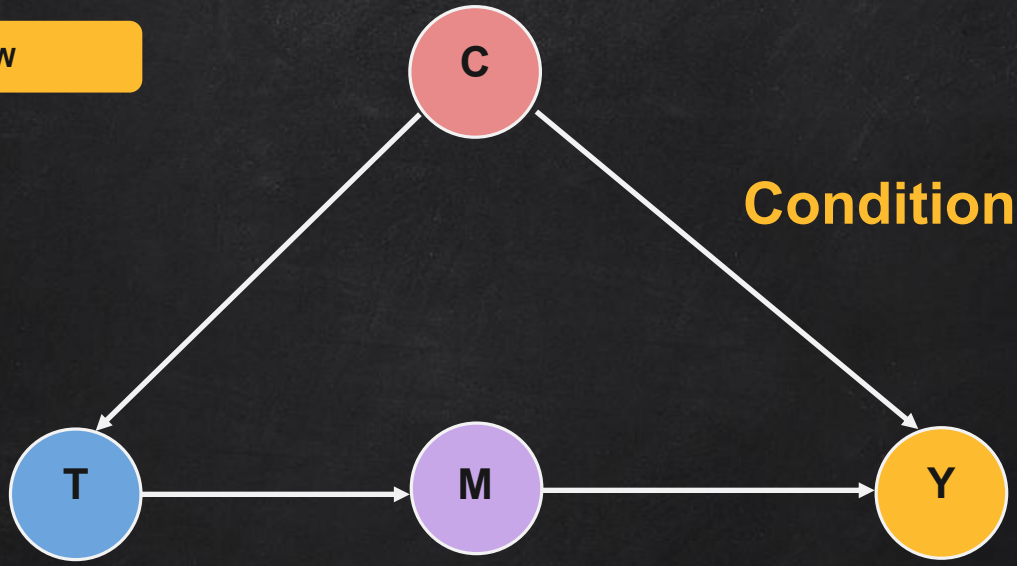
A practical d-Separation Algorithm

WORKFLOW

1. List every path between X and Y.
2. For each path, identify any collider(s).
3. Ask whether a non-collider on the path is conditioned on → BLOCKED.
4. If there is a collider, ask whether it or a descendant is conditioned on.
5. X and Y are d-separated only if ALL paths are blocked.

d-Separation: Exercise

WORKFLOW



Conditioning set: $S = \{C\}$

Question: $T \perp_D Y \mid C$?

d-Separation: Exercise 1

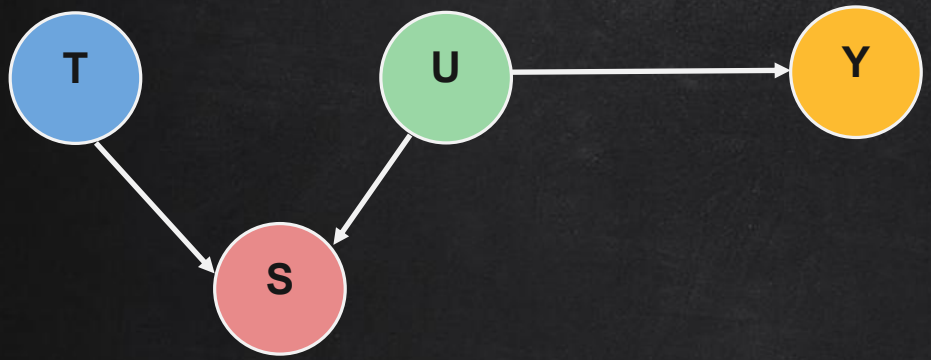
NO.

Path 1: $T \leftarrow C \rightarrow Y$ — blocked by conditioning on C

Path 2: $T \rightarrow M \rightarrow Y$ — still open

d-separation requires EVERY path to be blocked.

d-Separation: Exercise 2



Consider the path $T \rightarrow S \leftarrow U \rightarrow Y$

What happens if we condition on S?

Without conditioning on S

S is a collider.

T → S ← U → Y is BLOCKED at S.

This path does not transmit association.

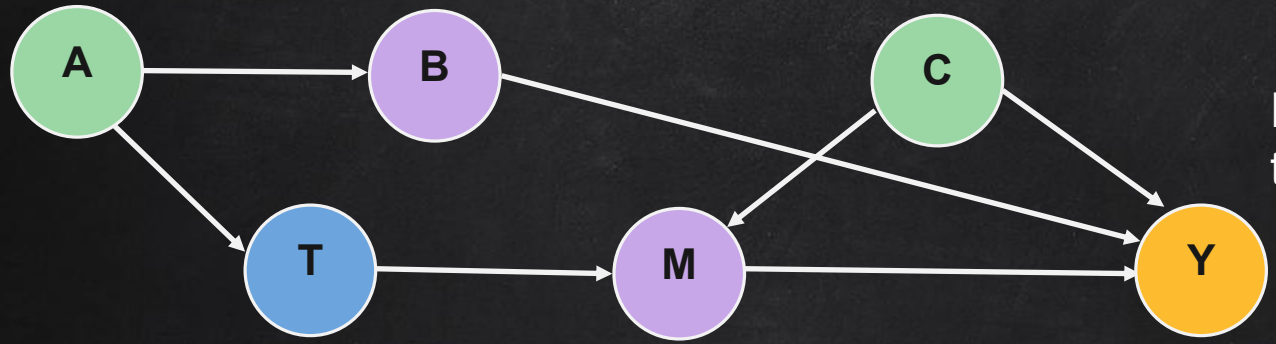
Condition on S

The collider is activated.

The path becomes OPEN.

T can now become associated with Y through U.

d-Separation: Exercise 3

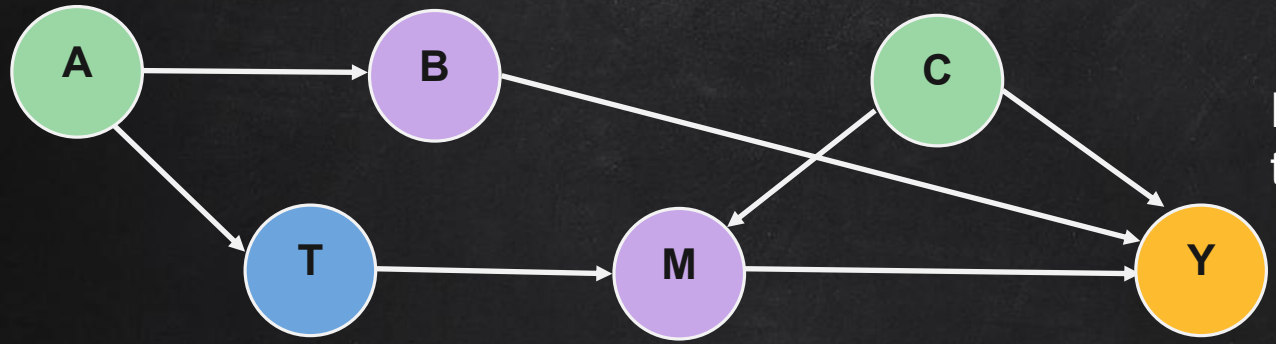


Find a conditioning set that d-separates T and Y.

Three paths:

- $T \rightarrow M \rightarrow Y$: Path 1 – Open, chain.
- $T \leftarrow A \rightarrow B \rightarrow Y$: Path 2 – Open, Backdoor path.
- $T \rightarrow M \leftarrow C \rightarrow Y$: Path 3 – Closed, M is a collider.

d-Separation: Exercise 3

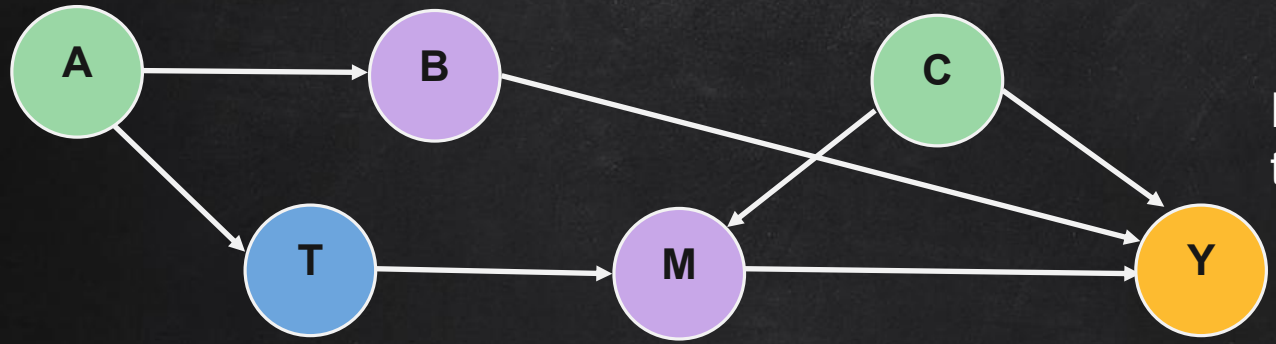


Find a conditioning set that d-separates T and Y.

Path 1 is an ordinary chain and is open by default.

The only internal node on this path is M. Therefor to block this path, we must condition on M. So any d-separation set should contain $\{M\}$.

d-Separation: Exercise 3



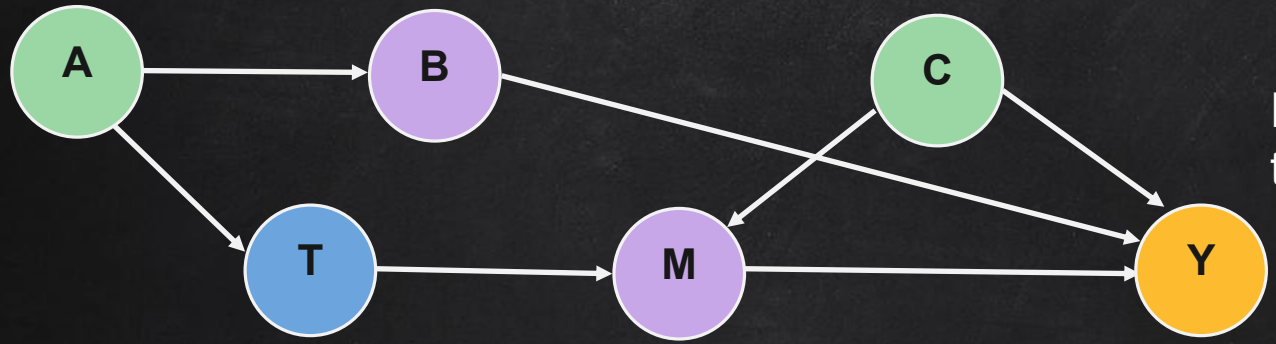
Find a conditioning set that d-separates T and Y.

Path 2 is also open by default. There are no colliders.

We can block it by conditioning on either A or B.

The conditioning set must also contain $\{A\}$ or $\{B\}$.

d-Separation: Exercise 3

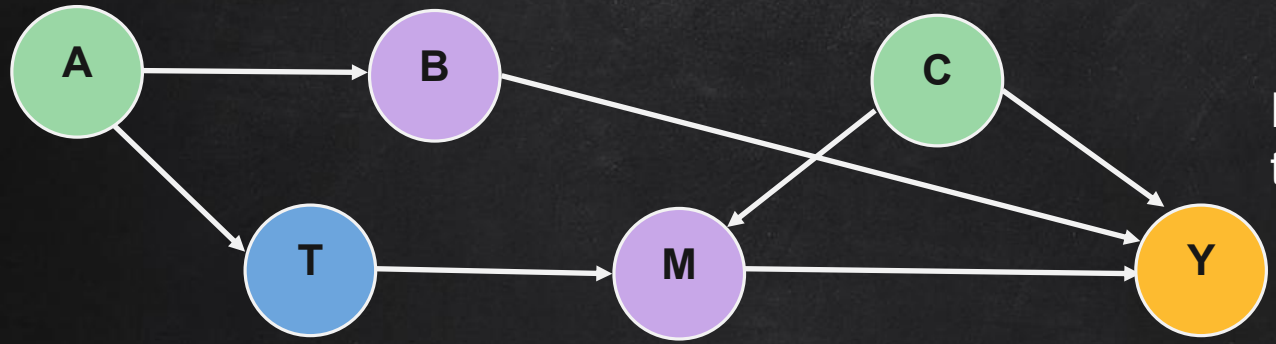


Find a conditioning set that d-separates T and Y.

So is $\{M, A\}$ enough?

Look at Path 3. $\{M\}$ is a collider, so conditioning on it opens Path 3. So we have to close it again. For that, we need to condition on $\{C\}$.

d-Separation: Exercise 3

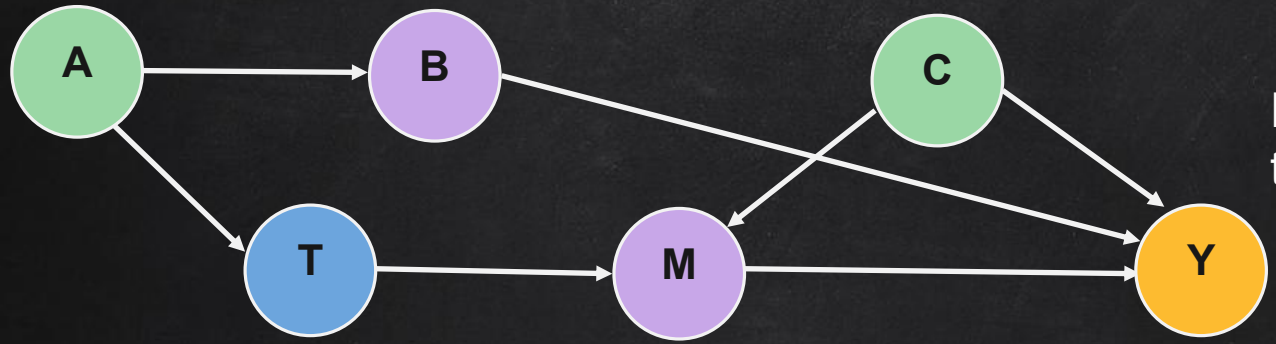


Find a conditioning set that d-separates T and Y.

$\{M\}$ is required for Path 1, $\{A\}$ or $\{B\}$ is required for Path 2. $\{C\}$ is required because conditioning on $\{M\}$ opens Path 3.

d-separation sets: $S = \{A, M, C\}$ or $\{B, M, C\}$.

d-Separation: Exercise 3



Find a conditioning set that d-separates T and Y.

T: Receiving an intense treatment.
Y: Recovery.
M: Biological response to treatment.
A: Baseline disease severity.
B: Baseline organ damage.
C: Genetic risk factor.

$$A \to T, \qquad A \to B \to Y$$

$$T \to M \to Y$$

$$C \to M, \qquad C \to Y$$

d-Separation ≠ Causal Adjustment

d-SEPARATION

Question:

Can we make T and Y conditionally independent?

Goal: BLOCK ALL paths.

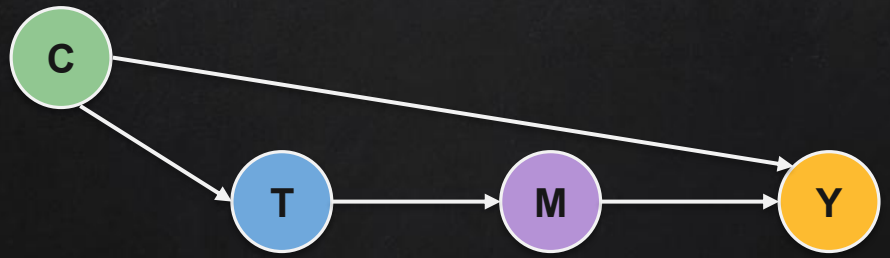
CAUSAL ADJUSTMENT: TOTAL EFFECT

Question:

What should we condition on to identify the total effect?

Goal: BLOCK NONCAUSAL backdoor paths.

Example:

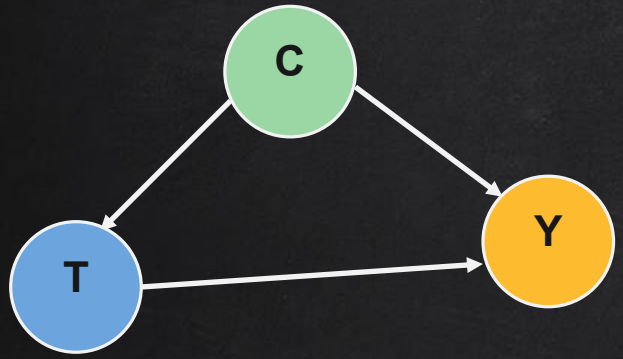


Backdoor path:
T ← C → Y

Causal path:
T → M → Y

For total effect: adjust for C, not M.

Backdoor path

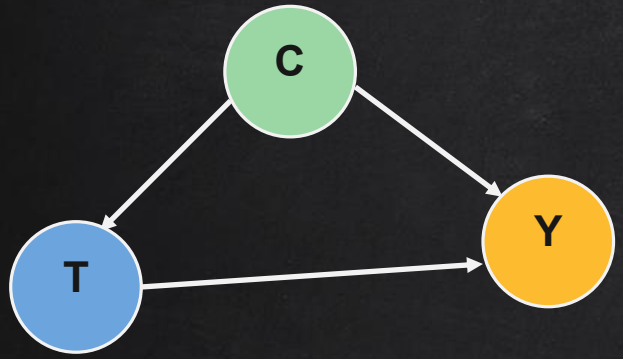


A path from T to Y that begins with an arrow INTO T.

- Choose a set S that blocks every backdoor path from T to Y.
- Do not include descendants of T in a standard total-effect adjustment set.
- After blocking backdoor paths, the remaining association from T to Y can be interpreted causally under the graph's assumptions.

Block noncausal paths. Leave causal paths open.

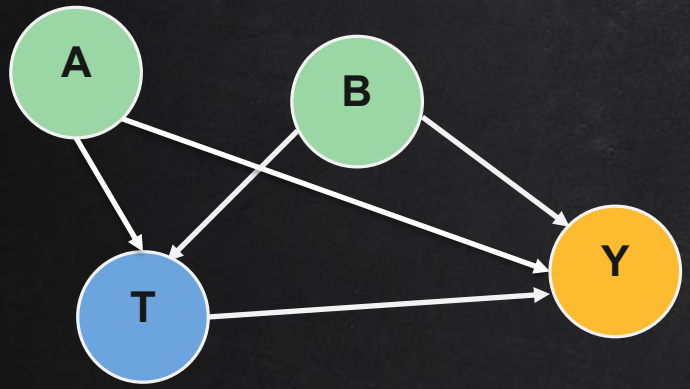
Backdoor path



Backdoor path: $T \leftarrow C \rightarrow Y$

Conditioning on C blocks the backdoor path without blocking $T \rightarrow Y$.

Backdoor path



Backdoor paths

$$T \leftarrow A \rightarrow Y \quad \text{and} \quad T \leftarrow B \rightarrow Y$$

Need to block BOTH paths: adjust for A and B.

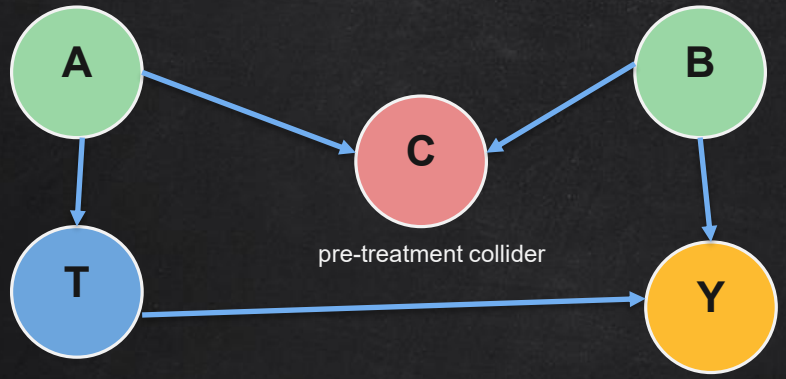
Minimal Adjustment Set

A valid adjustment set is MINIMAL if removing any variable from it would leave at least one backdoor path open.

Valid set
{Age, Severity, Sex}

Minimal?
Depends on the graph

“Pre-treatment” does NOT mean “adjust for it”

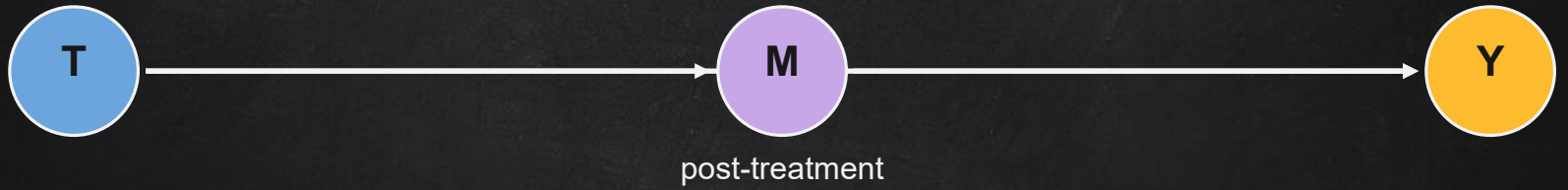


C occurs before treatment—but adjusting for C opens $T \leftarrow A \rightarrow C \leftarrow B \rightarrow Y$.

Example:

- T = applicant attended an expensive private high school
- Y= admission to a selective university
- A= family wealth
- B= parental education
- C= participation in an elite pre-college enrichment program

Post-Treatment variables



M is caused by T.

If estimating the TOTAL effect of T on Y, adjusting for M blocks part of the effect.

Mediator: part of the causal mechanism

MEDIATION



Total effect = direct pathway + mediated pathway.

Adjusting for M changes the estimand.

Total Effect vs Direct Effect

TOTAL EFFECT

Question:
What happens to Y if we intervene on T?

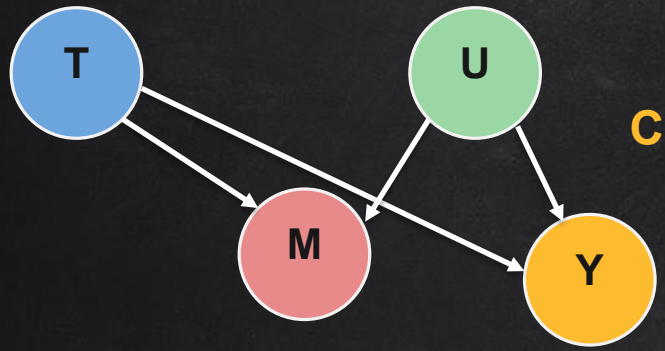
Do NOT block causal mediators when the total effect is the target.

DIRECT / MEDIATED EFFECTS

Question:
How much of the effect operates through M?

This is a different estimand and requires mediation-specific assumptions.

Collider adjustment bias with randomized T



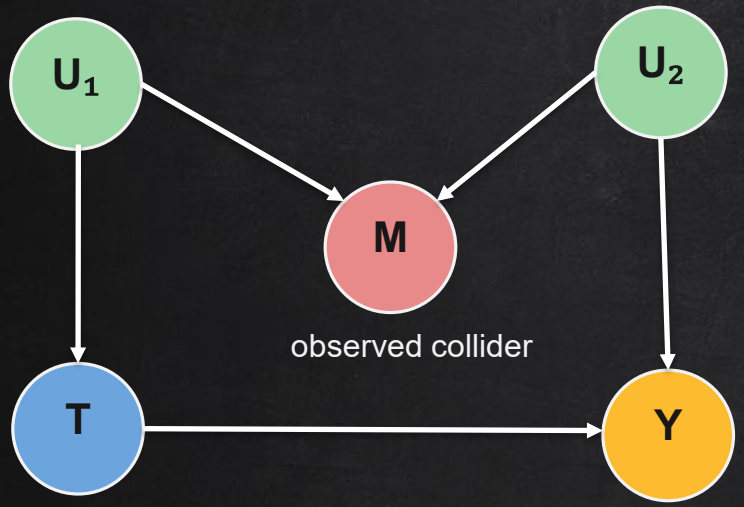
Conditioning on M opens: $\mathrm{T} \rightarrow \mathrm{M} \leftarrow \mathrm{U} \rightarrow \mathrm{Y}$

Example:

- T: randomized chemotherapy
- M: treatment-induced weight loss
- U: underlying frailty
- Y: survival

If you adjust for weight loss M, you induce an association between treatment and frailty, and that can bias the treatment effect on survival.
Randomization protects you at baseline, but you can destroy that protection by conditioning on the wrong post-treatment variable.

M-bias



Without adjusting for M, the path
$T \leftarrow U_1 \rightarrow M \leftarrow U_2 \rightarrow Y$ is blocked at M.

- Not every predictor of treatment or outcome is a confounder.
- A variable can be statistically associated with both T and Y yet still be a bad adjustment variable.
- Real-world covariate choice should combine subject-matter knowledge, temporal ordering, and causal structure.

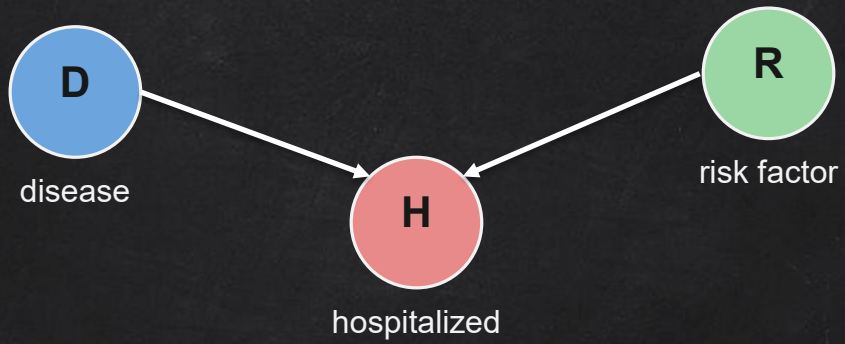
Overadjustment Bias

Overadjustment usually means conditioning on a variable that lies on, or is caused by, a causal pathway you intend to estimate.



Adjust for M → estimated "effect of T" no longer contains T → M → Y.

Selection Bias



If your dataset contains only H = 1 patients, you have conditioned on a collider.

Inside the hospital, D and R can appear associated even if they are independent in the population.

Selection can happen before analysis starts

- Eligibility criteria can condition on a collider.
- Complete-case analysis can condition on causes of missingness.
- Survey response can depend on both exposure and outcome-related factors.
- Clinic-based samples can differ systematically from the source population.

Ask: “What had to happen for someone to enter my dataset?”

How to draw a DAG from a research question

1. Define treatment/exposure T and outcome Y clearly.

2. List causes of T, causes of Y, and common causes.

3. Add plausible mediators and important selection variables.

4. Respect temporal ordering whenever possible.

5. Draw arrows from cause to consequence, not from correlation to correlation.

6. Use the graph to inspect backdoor paths and identify adjustment sets.

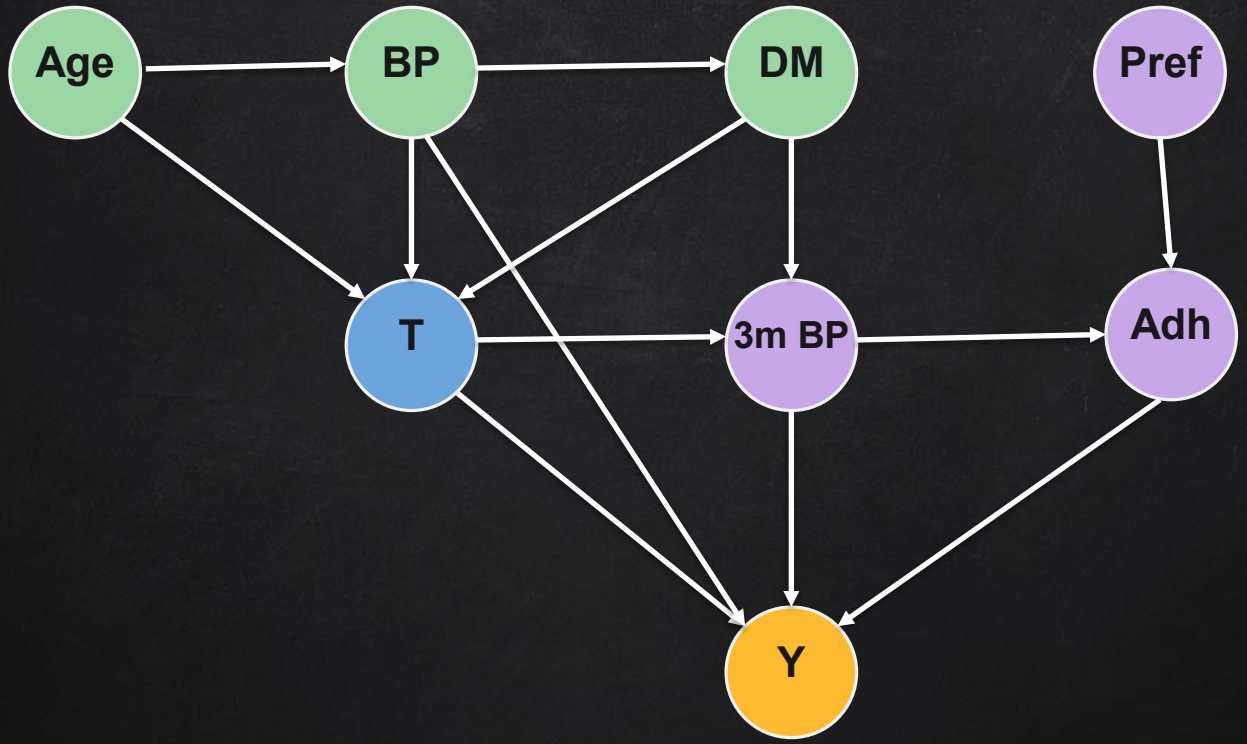
Build a graph

Research question: Does a new intensive blood-pressure treatment reduce 1-year cardiovascular events?

Candidate variables:
Age • baseline systolic BP • diabetes • physician treatment preference • treatment assignment • achieved BP at 3 months • medication adherence • cardiovascular event

Draw plausible DAG. Circle the variables you would adjust for to estimate the TOTAL treatment effect.

One plausible DAG



For TOTAL effect, what would we adjust for?

GOOD CANDIDATES

Baseline causes of treatment and outcome:

- baseline BP
- diabetes
- age (if it causes outcome and/or treatment through measured paths)
- physician preference if it lies on an open backdoor path between treatment and outcome.

DO NOT AUTOMATICALLY ADJUST

Post-treatment variables:

- achieved 3-month BP
- adherence

These may mediate the treatment effect or become colliders.

Next: Identification to Estimation

Today: Which variables should we adjust for?

Next: How do we create a fair treated-versus-untreated comparison?

PROPENSITY SCORE

$e(X) = P(T = 1 \mid X)$

Summarize treatment assignment using baseline covariates.

MATCHING & WEIGHTING

Build a comparison population in which treated and untreated groups are more comparable.

DIAGNOSTICS

Overlap, positivity, covariate balance, and what to do when comparable patients do not exist.

Preview question: Can an observational study be reweighted to mimic some features of randomization?